

Brutal Practice Paper B22 - Heat / Nutrition in Plants / Rational Numbers / Data Handling

Each question has 6 to 8 options; exactly ONE is correct. Answers are scattered - there is NO pattern. Every question needs several steps - read carefully and work it out fully.

1. A clinical thermometer reads 98.6 degrees Fahrenheit. Using $C = \frac{5}{9} \times (F - 32)$, what is this in Celsius, and is it normal body temperature? [\[Heat\]](#)
 - (A) 38 C, a mild fever
 - (B) 42 C, dangerously high
 - (C) 37 C, and this is normal body temperature
 - (D) 37 C, this is a high fever
 - (E) 36 C, slightly below normal
 - (F) 66.6 C, normal
 - (G) 40 C, normal

2. Find the sum of $-\frac{3}{4}$ and $\frac{5}{6}$, then subtract $-\frac{1}{3}$ from that result. What do you get? [\[Rational Numbers\]](#)
 - (A) $-\frac{1}{12}$
 - (B) $\frac{7}{12}$
 - (C) $-\frac{5}{12}$
 - (D) $\frac{5}{12}$
 - (E) $\frac{1}{12}$
 - (F) $\frac{11}{12}$

3. A leaf makes 12 g of glucose during 8 hours of steady sunlight. At the same rate per leaf, how much glucose would 20 such leaves make in 5 hours? [\[Nutrition in Plants\]](#)
 - (A) 75 g
 - (B) 60 g
 - (C) 240 g
 - (D) 300 g
 - (E) 120 g
 - (F) 150 g
 - (G) 96 g

4. The mean of six numbers is 24. Five of them are 18, 30, 27, 21 and 26. What is the sixth number? [\[Data Handling\]](#)
 - (A) 26
 - (B) 20
 - (C) 144
 - (D) 22
 - (E) 122
 - (F) 24
 - (G) 28

5. A beaker of hot water cools by 8 C in the first minute, and each later minute it cools by 2 C less than the minute before. If it started at 90 C, what is its temperature at the end of 4 minutes? [\[Heat\]](#)
- (A) 74 C
 - (B) 70 C
 - (C) 50 C
 - (D) 62 C
 - (E) 58 C
 - (F) 72 C
 - (G) 66 C
6. Which is the greater rational number, $-7/9$ or $-5/6$? [\[Rational Numbers\]](#)
- (A) $-12/15$
 - (B) cannot be compared
 - (C) $-5/6$
 - (D) $-35/45$ only
 - (E) they are equal
 - (F) $-7/9$
7. A leaf has about 300 stomata on every square millimetre of its underside. About how many stomata are on the underside of a leaf with an area of 4 square centimetres? [\[Nutrition in Plants\]](#)
- (A) 1,20,000
 - (B) 12,000
 - (C) 4800
 - (D) 1,200,000
 - (E) 30,000
 - (F) 12,00,000
 - (G) 1200
8. The midday temperatures (Celsius) for six days were 23, 17, 31, 12, 28 and 19. Find the median and the range. [\[Data Handling\]](#)
- (A) median 22, range 12
 - (B) median 21.5, range 19
 - (C) median 21, range 19
 - (D) median 19, range 19
 - (E) median 20, range 18
 - (F) median 21, range 31
 - (G) median 23, range 19
9. A laboratory thermometer is marked from -10 C to 110 C, with the smallest marks 1 C apart. How many degrees does the whole scale span, and how many 5 C gaps fit along it? [\[Heat\]](#)
- (A) span 110 C, 22 gaps
 - (B) span 130 C, 26 gaps
 - (C) span 120 C, 22 gaps
 - (D) span 100 C, 20 gaps
 - (E) span 120 C, 25 gaps
 - (F) span 120 C, 24 gaps
 - (G) span 100 C, 24 gaps

10. Evaluate the product $(-2/3) \times (9/10) \times (-5/6)$. Give the answer in lowest terms. [\[Rational Numbers\]](#)

- (A) $-1/2$
- (B) $90/180$ only
- (C) $-3/10$
- (D) $3/10$
- (E) $1/2$
- (F) $1/4$
- (G) $2/3$

11. In bright light a water plant releases 30 oxygen bubbles each minute. When the light is dimmed the rate falls to $2/5$ of this. How many bubbles does it release in 6 minutes in the dim light? [\[Nutrition in Plants\]](#)

- (A) 60 bubbles
- (B) 72 bubbles
- (C) 12 bubbles
- (D) 180 bubbles
- (E) 120 bubbles
- (F) 36 bubbles
- (G) 90 bubbles

12. In a class survey, shoe sizes recorded were 5, 6, 7, 6, 8, 6, 7, 5 and 6. What is the modal shoe size, and how many students wore it? [\[Data Handling\]](#)

- (A) size 8, worn by 1 student
- (B) size 6, worn by 3 students
- (C) size 6, worn by 4 students
- (D) size 7, worn by 2 students
- (E) size 7, worn by 4 students
- (F) size 5, worn by 2 students
- (G) size 6, worn by 9 students

13. Water at 25 C is heated so its temperature rises a steady 5 C every minute. At sea level water boils at 100 C. After how many minutes does it reach boiling, and how many degrees will it have risen? [\[Heat\]](#)

- (A) 20 minutes, a rise of 100 C
- (B) 25 minutes, a rise of 125 C
- (C) 15 minutes, a rise of 100 C
- (D) 15 minutes, a rise of 75 C
- (E) 12 minutes, a rise of 60 C
- (F) 14 minutes, a rise of 70 C
- (G) 19 minutes, a rise of 95 C

14. Divide $-8/15$ by $-4/5$. Give the result in lowest terms. [\[Rational Numbers\]](#)

- (A) $3/2$
- (B) $32/75$
- (C) $-32/75$
- (D) $2/5$
- (E) $2/3$
- (F) $8/12$
- (G) $-2/3$

15. A green plant needs sunlight, chlorophyll, carbon dioxide and water to photosynthesise. Of these four, which two actually become part of the glucose that is built, while the other two only make the reaction happen? [\[Nutrition in Plants\]](#)

- (A) all four become part of the glucose
- (B) only water becomes part of the glucose
- (C) sunlight and chlorophyll become part of the glucose
- (D) carbon dioxide and water become part of the glucose; sunlight and chlorophyll only enable it
- (E) water and sunlight become part of the glucose
- (F) carbon dioxide and chlorophyll become part of the glucose
- (G) only carbon dioxide becomes part of the glucose

16. A bag holds 5 red, 3 blue and 2 green balls, all the same size. One ball is drawn without looking. What is the probability that it is blue? [\[Data Handling\]](#)

- (A) $\frac{3}{5}$
- (B) $\frac{7}{10}$
- (C) $\frac{3}{10}$
- (D) $\frac{2}{10}$
- (E) $\frac{3}{7}$
- (F) $\frac{1}{3}$
- (G) $\frac{5}{10}$

17. A metal spoon and a wooden spoon are left standing in hot soup. After two minutes the metal handle is at 55 C and the wooden handle at 31 C, while the room is at 28 C. Which conducts heat better, and how many degrees did each handle rise above the room? [\[Heat\]](#)

- (A) metal; metal rose 24 C, wood rose 3 C
- (B) wood; metal rose 27 C, wood rose 3 C
- (C) metal; metal rose 27 C, wood rose 3 C
- (D) wood; metal rose 3 C, wood rose 27 C
- (E) metal; both rose by the same amount
- (F) metal; metal rose 23 C, wood rose 27 C
- (G) metal; metal rose 55 C, wood rose 31 C

18. Find the rational number that lies exactly halfway between $\frac{2}{3}$ and $\frac{3}{4}$. [\[Rational Numbers\]](#)

- (A) $\frac{17}{12}$
- (B) $\frac{7}{12}$
- (C) $\frac{17}{24}$
- (D) $\frac{17}{48}$
- (E) $\frac{5}{12}$
- (F) $\frac{1}{2}$
- (G) $\frac{5}{7}$

- 19.** A farmer grows a crop of peas (a legume) and then a crop of wheat in the same field. Why does the pea crop help the wheat that follows, and which tiny helpers in the pea roots are responsible? [\[Nutrition in Plants\]](#)
- (A) the peas add carbon dioxide to the soil through their roots
 - (B) the peas leave the soil richer in nitrogen, thanks to Rhizobium bacteria in their root nodules
 - (C) the peas remove all nitrogen, so wheat grows in clean soil
 - (D) the Rhizobium bacteria eat the wheat's pests
 - (E) the peas make the soil acidic, which wheat prefers
 - (F) the peas leave the soil richer in nitrogen, thanks to fungi on the leaves
 - (G) the peas store sunlight in the soil for the wheat
- 20.** Rohan rolls one fair die whose faces show 1 to 6. Find the probability that his score is more than 4. [\[Data Handling\]](#)
- (A) $\frac{1}{4}$
 - (B) $\frac{1}{3}$
 - (C) $\frac{4}{6}$
 - (D) $\frac{5}{6}$
 - (E) $\frac{1}{6}$
 - (F) $\frac{2}{3}$
 - (G) $\frac{1}{2}$
- 21.** On a hot afternoon at the coast the land becomes much hotter than the sea. The air above the land warms, becomes lighter and rises, and cooler air moves in to take its place. In which direction does the breeze blow, and what is it called? [\[Heat\]](#)
- (A) there is no breeze in the afternoon
 - (B) from the land towards the sea; a land breeze
 - (C) from the sea towards the land; a sea breeze
 - (D) from the land towards the sea; a sea breeze
 - (E) from the sea towards the land; a land breeze
 - (F) straight upwards; a convection breeze
 - (G) from the sea towards the land; a monsoon
- 22.** Write $-36 / (-48)$ in standard form, that is, in lowest terms with a positive denominator. [\[Rational Numbers\]](#)
- (A) $\frac{9}{12}$
 - (B) $\frac{3}{4}$
 - (C) $\frac{4}{3}$
 - (D) $\frac{36}{48}$
 - (E) $\frac{6}{8}$
 - (F) $-\frac{4}{3}$
 - (G) $-\frac{3}{4}$
- 23.** A whole tree carries about 8000 leaves, and each leaf makes on average 0.5 g of glucose per day. About how many kilograms of glucose does the tree make over a 30-day month? [\[Nutrition in Plants\]](#)
- (A) 60 kg
 - (B) 240 kg
 - (C) 120 kg
 - (D) 40 kg
 - (E) 4 kg
 - (F) 12 kg
 - (G) 1200 kg

- 24.** The mean age of five players in a team is 12 years. A new player aged 18 joins the team. What is the new mean age of the six players? [\[Data Handling\]](#)
- (A) 12 years
 - (B) 13 years
 - (C) 12.5 years
 - (D) 13.5 years
 - (E) 15 years
 - (F) 30 years
- 25.** Two identical bottles of water, one painted black and one painted white, both start at 30 C in sunshine. After an hour the black one is at 48 C and the white one at 38 C. Which absorbed more heat, and how many times as large was its temperature rise? [\[Heat\]](#)
- (A) both rose by the same amount
 - (B) the white bottle; 2.25 times as large a rise
 - (C) the black bottle; 1.26 times as large a rise
 - (D) the black bottle; 2.25 times as large a rise
 - (E) the black bottle; 10 times as large a rise
 - (F) the black bottle; 2 times as large a rise
- 26.** The product of a certain rational number and $-\frac{5}{8}$ is $\frac{15}{16}$. What is the rational number? [\[Rational Numbers\]](#)
- (A) $-\frac{3}{8}$
 - (B) $\frac{2}{3}$
 - (C) $-\frac{3}{2}$
 - (D) $-\frac{2}{3}$
 - (E) $\frac{75}{128}$
 - (F) $\frac{3}{2}$
- 27.** A variegated leaf (green only at its edges, white in the middle) is kept in sunlight, then tested with iodine for starch. Which parts turn blue-black, and what does this prove about photosynthesis? [\[Nutrition in Plants\]](#)
- (A) the whole leaf turns blue-black, proving light alone makes food
 - (B) only the green edges turn blue-black, proving chlorophyll is needed to make food
 - (C) only the white middle turns blue-black, proving white parts make food
 - (D) only the green edges turn red, proving sugar not starch
 - (E) no part turns blue-black, proving leaves cannot make starch
 - (F) the white middle turns blue, the green stays pale
- 28.** The mean of ten observations was found to be 15. Later one observation was discovered to have been read as 12 instead of the correct 21. What is the corrected mean? [\[Data Handling\]](#)
- (A) 16
 - (B) 159
 - (C) 15
 - (D) 15.09
 - (E) 16.5
 - (F) 15.9
 - (G) 14.1

- 29.** A clinical thermometer runs from 35 C to 42 C and its smallest marks are 0.1 C apart. How many small divisions are printed along its scale? [\[Heat\]](#)
- (A) 70 divisions
 - (B) 700 divisions
 - (C) 35 divisions
 - (D) 71 divisions
 - (E) 42 divisions
 - (F) 69 divisions
 - (G) 7 divisions
- 30.** A diver starts at the surface, goes down $\frac{3}{5}$ km, then rises $\frac{1}{4}$ km, then goes down a further $\frac{7}{20}$ km. Taking below the surface as negative, what is his final position? [\[Rational Numbers\]](#)
- (A) $-\frac{3}{10}$ km
 - (B) $\frac{7}{10}$ km
 - (C) $-\frac{7}{10}$ km (below the surface)
 - (D) $-\frac{2}{5}$ km
 - (E) $-\frac{9}{10}$ km
 - (F) $-\frac{1}{2}$ km
 - (G) $-\frac{13}{10}$ km
- 31.** A green leaf of area 50 square centimetres makes 0.2 g of glucose for every 10 square centimetres of leaf each hour. How much glucose does the whole leaf make in 3 hours? [\[Nutrition in Plants\]](#)
- (A) 3 g
 - (B) 0.3 g
 - (C) 0.6 g
 - (D) 1.5 g
 - (E) 30 g
 - (F) 1 g
- 32.** Seven students scored 40, 55, 55, 70, 80, 55 and 65 in a test. By how much does the mean exceed the mode? [\[Data Handling\]](#)
- (A) 5
 - (B) 15
 - (C) 60
 - (D) 20
 - (E) 55
 - (F) 0
 - (G) 10
- 33.** A child's fever reads 104 degrees Fahrenheit. Using $C = \frac{5}{9} \times (F - 32)$, what is the temperature in Celsius, and how far is it above the normal 37 C? [\[Heat\]](#)
- (A) 40 C, which is 4 C above normal
 - (B) 39 C, which is 2 C above normal
 - (C) 42 C, which is 5 C above normal
 - (D) 104 C, far above normal
 - (E) 40 C, which is 3 C above normal
 - (F) 58 C, which is 21 C above normal
 - (G) 37 C, which is normal

34. Arrange these rational numbers in ascending order: $-\frac{2}{3}$, $\frac{3}{-4}$, $-\frac{1}{2}$, $\frac{5}{-6}$. [\[Rational Numbers\]](#)

- (A) $-\frac{2}{3}$, $-\frac{3}{4}$, $-\frac{1}{2}$, $-\frac{5}{6}$
- (B) $-\frac{5}{6}$, $-\frac{2}{3}$, $-\frac{3}{4}$, $-\frac{1}{2}$
- (C) $-\frac{1}{2}$, $-\frac{3}{4}$, $-\frac{2}{3}$, $-\frac{5}{6}$
- (D) $-\frac{1}{2}$, $-\frac{2}{3}$, $-\frac{3}{4}$, $-\frac{5}{6}$
- (E) $-\frac{5}{6}$, $-\frac{3}{4}$, $-\frac{2}{3}$, $-\frac{1}{2}$
- (F) $-\frac{3}{4}$, $-\frac{5}{6}$, $-\frac{2}{3}$, $-\frac{1}{2}$
- (G) $-\frac{2}{3}$, $-\frac{1}{2}$, $-\frac{3}{4}$, $-\frac{5}{6}$

35. Cuscuta (amarbel) is a pale, leafless climber that twines over other plants and sends tiny suckers into their stems. How does it feed, and what is this feeding relationship called? [\[Nutrition in Plants\]](#)

- (A) it draws food from the host; it is a saprotroph
- (B) it draws ready-made food from the host plant; it is a parasite
- (C) it absorbs food from dead leaves; it is a parasite
- (D) it lives on sunlight alone; it is a parasite
- (E) it makes its own food by photosynthesis; it is an autotroph
- (F) it traps insects for food; it is insectivorous

36. A batsman's runs in eight innings were 45, 0, 72, 12, 0, 88, 36 and 0. Find the mode and the median of his scores. [\[Data Handling\]](#)

- (A) mode 0, median 45
- (B) mode 36, median 24
- (C) mode 0, median 36
- (D) mode 0, median 12
- (E) mode 0, median 31
- (F) mode 12, median 24
- (G) mode 0, median 24

37. A room heater warms the air near it. To heat the whole room most evenly by convection, where is it best placed, because warm air does what? [\[Heat\]](#)

- (A) near the ceiling, because warm air sinks
- (B) near the floor, because warm air rises and cool air sinks to replace it
- (C) near the floor, because warm air does not move at all
- (D) in the exact centre at head height, because air does not move
- (E) near the ceiling, because cool air rises
- (F) near the floor, because warm air sinks and stays low
- (G) anywhere, because heat spreads only by conduction

38. How many rational numbers lie between $-\frac{1}{2}$ and $\frac{1}{2}$? [\[Rational Numbers\]](#)

- (A) only one, which is 0
- (B) infinitely many
- (C) exactly nine
- (D) exactly two
- (E) none at all
- (F) exactly 0, since 0 is not rational
- (G) exactly four

- 39.** A pitcher plant grows in marshy soil that is very poor in one nutrient. It traps and digests insects to get that nutrient. Which nutrient is it short of, and why can it still make its own glucose? [\[Nutrition in Plants\]](#)
- (A) it is short of nitrogen; it gets glucose from the insects
 - (B) it is short of carbon; it cannot make glucose at all
 - (C) it is short of nitrogen; it still makes glucose because it has green leaves with chlorophyll
 - (D) it is short of nitrogen; it has no chlorophyll
 - (E) it is short of sunlight; insects supply the energy
 - (F) it is short of oxygen; the insects breathe for it
- 40.** A game wheel is divided into 8 equal sectors marked 1 through 8, and the pointer is spun once. Find the chance it stops on a multiple of 3. [\[Data Handling\]](#)
- (A) $\frac{3}{4}$
 - (B) $\frac{1}{3}$
 - (C) $\frac{1}{2}$
 - (D) $\frac{1}{4}$
 - (E) $\frac{1}{8}$
 - (F) $\frac{2}{8}$ of nothing
 - (G) $\frac{3}{8}$
- 41.** Midday temperatures (in Celsius) over five days were 31, 34, 29, 36, 30. What was the mean midday temperature, and how many degrees above the mean was the hottest day? [\[Heat\]](#)
- (A) mean 32 C, hottest day 4 C above
 - (B) mean 32 C, hottest day 6 C above
 - (C) mean 34 C, hottest day 2 C above
 - (D) mean 32 C, hottest day 36 C above
 - (E) mean 31 C, hottest day 5 C above
 - (F) mean 30 C, hottest day 6 C above
 - (G) mean 33 C, hottest day 3 C above
- 42.** Simplify $(\frac{3}{4} - \frac{5}{8})$ divided by $(-\frac{1}{8})$. Give the answer. [\[Rational Numbers\]](#)
- (A) -8
 - (B) -1
 - (C) $\frac{1}{64}$
 - (D) 8
 - (E) 1
 - (F) $-\frac{1}{8}$
 - (G) $-\frac{1}{64}$
- 43.** Bread left in a warm, damp place grows fuzzy fungus. The fungus has no chlorophyll. How does it feed on the bread, and what is this kind of feeder called? [\[Nutrition in Plants\]](#)
- (A) it bites pieces of bread like an animal; a parasite
 - (B) it makes its own food from sunlight; an autotroph
 - (C) it draws food from a living host; a parasite
 - (D) it releases juices that break down the bread, then absorbs the dissolved food; a saprotroph
 - (E) it traps tiny insects in the bread; insectivorous
 - (F) it shares food with the bread; a symbiont

44. The mean of four numbers is 9. Three of them are 6, 11 and 8. Find the fourth number, then find the new mean if that fourth number is doubled. [\[Data Handling\]](#)

- (A) fourth is 11, new mean is 22
- (B) fourth is 11, new mean is 11.75
- (C) fourth is 11, new mean is 14.5
- (D) fourth is 9, new mean is 11.75
- (E) fourth is 11, new mean is 12
- (F) fourth is 7, new mean is 10.5
- (G) fourth is 11, new mean is 9

45. Why do two thin sweaters worn together keep you warmer than a single thick sweater of the same total thickness? [\[Heat\]](#)

- (A) the inner sweater conducts your heat outwards faster
- (B) an extra layer of trapped air between them, and air is a poor conductor
- (C) the two sweaters are heavier, and heavier clothes are warmer
- (D) the outer sweater reflects sunlight onto you
- (E) two sweaters block the wind but trap no air
- (F) trapped air is a very good conductor of heat
- (G) wool makes heat by itself when doubled

46. Evaluate $-5/12 + 7/8 - 1/6$. Give the answer in lowest terms. [\[Rational Numbers\]](#)

- (A) $5/24$
- (B) $7/24$
- (C) $17/24$
- (D) $-7/24$
- (E) $11/24$
- (F) $1/24$

47. A lichen is really two organisms living together: a fungus and an alga. The alga makes food by photosynthesis; the fungus provides shelter, water and minerals. What is this give-and-take relationship called? [\[Nutrition in Plants\]](#)

- (A) symbiosis, where both partners benefit
- (B) insectivory, where they trap insects together
- (C) competition, where they fight for food
- (D) photosynthesis, done by the fungus
- (E) predation, where the fungus eats the alga
- (F) parasitism, where the fungus harms the alga

48. Class A has 20 students with a mean test score of 60. Class B has 30 students with a mean of 70. What is the mean score of all 50 students taken together? [\[Data Handling\]](#)

- (A) 68
- (B) 64
- (C) 63
- (D) 67
- (E) 130
- (F) 66
- (G) 65

- 49.** A copper rod and a glass rod of the same size are heated at one end. A drop of wax is stuck halfway along each. The wax on copper melts in 20 seconds; the wax on glass has not melted after 5 minutes. What does this show, and roughly how many times faster did heat travel along the copper? [\[Heat\]](#)
- (A) glass is the better conductor; about 15 times faster
 - (B) both conduct equally; the wax was different
 - (C) glass is the better conductor; about 3 times faster
 - (D) copper is the better conductor; heat reached the middle at least 15 times faster
 - (E) neither conducts; the wax melted by itself
 - (F) copper is the better conductor; exactly 4 times faster
 - (G) copper is the better conductor; about 5 times faster
- 50.** What number must be added to $-\frac{7}{12}$ to give $\frac{3}{4}$? [\[Rational Numbers\]](#)
- (A) $\frac{16}{12}$ unsimplified only
 - (B) $\frac{2}{12}$
 - (C) $\frac{4}{3}$
 - (D) $\frac{1}{3}$
 - (E) $\frac{5}{6}$
 - (F) $-\frac{4}{3}$
- 51.** In a starch-test experiment, a leaf is first boiled in water, then boiled in alcohol, then washed and tested with iodine. Why is the leaf boiled in alcohol before the iodine is added? [\[Nutrition in Plants\]](#)
- (A) to add starch to the leaf
 - (B) to add chlorophyll so the colour shows
 - (C) to kill insects on the leaf
 - (D) to wash off the iodine quickly
 - (E) to make the leaf green again
 - (F) to remove the green chlorophyll so the blue-black starch colour can be seen clearly
- 52.** A fair coin is tossed twice. What is the probability of getting two heads? [\[Data Handling\]](#)
- (A) $\frac{1}{4}$
 - (B) $\frac{3}{4}$
 - (C) $\frac{1}{3}$
 - (D) $\frac{2}{4}$ of three
 - (E) $\frac{1}{8}$
 - (F) $\frac{2}{3}$
- 53.** At night the land cools down faster than the sea, so the air above the sea stays warmer and rises. In which direction does the night breeze blow at the coast, and what is it called? [\[Heat\]](#)
- (A) straight downwards onto the sea; a sea breeze
 - (B) there is no breeze at night
 - (C) from the land towards the sea; a land breeze
 - (D) from the sea towards the land; a sea breeze
 - (E) from the land towards the sea; a monsoon
 - (F) from the land towards the sea; a sea breeze
 - (G) from the sea towards the land; a land breeze

54. What number must be subtracted from $-\frac{2}{5}$ to give $-\frac{9}{10}$? [\[Rational Numbers\]](#)

- (A) $-\frac{7}{10}$
- (B) $\frac{13}{10}$
- (C) $-\frac{13}{10}$
- (D) $-\frac{1}{2}$
- (E) $-\frac{1}{10}$
- (F) $\frac{1}{10}$
- (G) $\frac{1}{2}$

55. A 2400 cm-squared patch of pond is covered with floating green algae. Algae cover $\frac{5}{8}$ of the patch; the rest is open water. How many square centimetres are covered by algae, and how many are open water? [\[Nutrition in Plants\]](#)

- (A) 2000 cm-squared algae, 400 cm-squared open water
- (B) 1500 cm-squared algae, 800 cm-squared open water
- (C) 1600 cm-squared algae, 800 cm-squared open water
- (D) 1500 cm-squared algae, 900 cm-squared open water
- (E) 1500 cm-squared algae, 1000 cm-squared open water
- (F) 1200 cm-squared algae, 1200 cm-squared open water
- (G) 900 cm-squared algae, 1500 cm-squared open water

56. Over six days a shop's temperatures (Celsius) were 28, 31, 29, 33, 30 and an unknown reading. The six-day mean was exactly 30. What was the unknown reading? [\[Data Handling\]](#)

- (A) 151
- (B) 28
- (C) 180
- (D) 30
- (E) 27
- (F) 29
- (G) 31

57. A swimming pool's water is at 26 C in the morning. By afternoon it has warmed by 15% of its morning temperature. What is the afternoon temperature? [\[Heat\]](#)

- (A) 30.9 C
- (B) 26.15 C
- (C) 41 C
- (D) 29.9 C
- (E) 29.6 C
- (F) 31.5 C
- (G) 39 C

58. By what number should $-\frac{5}{9}$ be multiplied to give 1? [\[Rational Numbers\]](#)

- (A) $-\frac{45}{45}$
- (B) $\frac{1}{5}$
- (C) $-\frac{9}{5}$
- (D) $-\frac{5}{9}$
- (E) $\frac{5}{9}$
- (F) -1

59. During photosynthesis a plant takes in carbon dioxide and gives out oxygen. During respiration the same plant takes in oxygen and gives out carbon dioxide. On a bright sunny day a plant photosynthesises much faster than it respire. Overall, which gas does it give out, and which does it take in? [\[Nutrition in Plants\]](#)

- (A) it only respire in daylight, giving out carbon dioxide
- (B) overall it gives out carbon dioxide and takes in oxygen
- (C) it gives out oxygen and also takes in oxygen
- (D) it stops both processes in bright light
- (E) it gives out nitrogen and takes in oxygen
- (F) it gives out and takes in nothing on a sunny day
- (G) overall it gives out oxygen and takes in carbon dioxide

60. A box contains tickets numbered 1 to 20. One ticket is drawn at random. What is the probability that the number is a multiple of 5 or a multiple of 4 (counting any number that is both only once)? [\[Data Handling\]](#)

- (A) $\frac{6}{20}$
- (B) $\frac{1}{2}$
- (C) $\frac{8}{20} = \frac{2}{5}$
- (D) $\frac{9}{20}$
- (E) $\frac{1}{4}$
- (F) $\frac{10}{20}$

61. Heat can travel by conduction, convection or radiation. The Sun's heat reaches the Earth across empty space where there is no air or matter. By which method does it travel, and why not the others? [\[Heat\]](#)

- (A) conduction, because space conducts heat directly
- (B) radiation, because space is full of air to carry heat
- (C) all three equally, because space allows everything
- (D) radiation, because conduction and convection both need a material to carry the heat
- (E) conduction, because the Sun touches the Earth
- (F) convection, because there is no matter in space
- (G) convection, because hot air rises through space

62. If $\frac{2}{3}$ of a number is $-\frac{8}{9}$, what is the number? [\[Rational Numbers\]](#)

- (A) $\frac{3}{4}$
- (B) $-\frac{16}{27}$
- (C) $\frac{4}{3}$
- (D) $-\frac{4}{3}$
- (E) $\frac{16}{27}$
- (F) $-\frac{3}{4}$

63. A scientist covers a healthy potted plant with a bell jar that also holds a dish of caustic potash, which soaks up all the carbon dioxide inside. After a day in sunlight a leaf is starch-tested. What is the result, and what does it prove? [\[Nutrition in Plants\]](#)

- (A) starch forms only in the dark, proving light stops it
- (B) no starch forms, proving chlorophyll was removed
- (C) no starch forms, proving carbon dioxide is needed for photosynthesis
- (D) plenty of starch forms, proving CO_2 is not needed
- (E) the leaf dies at once from lack of water
- (F) starch forms, proving the caustic potash adds food
- (G) no starch forms, proving sunlight is not needed

64. The five numbers 5, 8, x, 14 and 17 are written in ascending order and their median is 11. What is x?

[Data Handling]

- (A) middle of 5 and 17
- (B) 9
- (C) 12
- (D) 11
- (E) 8
- (F) 10
- (G) 13.5

65. A thermos flask keeps a 90 C drink hot. It has a shiny silvered inner wall and a vacuum gap between its two walls. The vacuum and the shiny coating mainly stop heat loss by which two methods? [Heat]

- (A) the vacuum stops conduction and convection; the shiny coating cuts radiation
- (B) the shiny coating stops conduction; the vacuum stops radiation
- (C) the glass walls stop all three by themselves
- (D) the vacuum stops only conduction; the coating stops convection
- (E) the vacuum stops radiation; the shiny coating stops conduction
- (F) the vacuum adds heat; the coating reflects sound
- (G) the cork lid stops radiation; the vacuum stops nothing

66. On the number line, which of these rational numbers lies immediately to the LEFT of $-\frac{3}{5}$ (that is, is just smaller than it)? [Rational Numbers]

- (A) $-\frac{4}{5}$
- (B) $-\frac{1}{2}$
- (C) 0
- (D) $-\frac{1}{5}$
- (E) $-\frac{2}{5}$
- (F) $-\frac{1}{2}$ and $-\frac{2}{5}$ both

67. Out of every 100 sugar molecules a leaf makes, the plant uses 35 at once for its own energy and stores the rest as starch. If a leaf makes 800 sugar molecules in a day, how many are stored as starch?

[Nutrition in Plants]

- (A) 560 molecules
- (B) 520 molecules
- (C) 260 molecules
- (D) 280 molecules
- (E) 320 molecules
- (F) 800 molecules

68. A double bar graph shows a shop's sales: in week 1 it sold 40 shirts and 25 trousers; in week 2, 55 shirts and 35 trousers. By how many items did the total weekly sales rise from week 1 to week 2? [Data Handling]

[Data Handling]

- (A) 20
- (B) 90
- (C) 15
- (D) 10
- (E) 25
- (F) 65

69. An iron ball is heated until it just fits too tightly to pass through a metal ring it slipped through when cold. Heating made the ball do what, and to let it pass through again you should do what to the ring? [\[Heat\]](#)

- (A) the ring shrank by itself; do nothing
- (B) the ball shrank; heat the ring
- (C) the ball expanded; squeeze the ring smaller
- (D) the ball expanded; heat the ring so it expands and its hole widens
- (E) the ball melted; let it solidify in the ring
- (F) the ball shrank; cool the ring
- (G) the ball expanded; cool the ring so its hole widens

70. Which of these rational numbers is equal to $-\frac{4}{7}$? [\[Rational Numbers\]](#)

- (A) $-\frac{12}{21}$
- (B) $-\frac{8}{21}$
- (C) $\frac{12}{21}$
- (D) $-\frac{7}{4}$
- (E) $-\frac{4}{14}$
- (F) $\frac{4}{7}$
- (G) $-\frac{11}{14}$

71. Stomata are tiny pores, each bordered by two guard cells, that open and close to control gas exchange. On a hot, dry afternoon a plant tends to close its stomata. What is the main benefit, and what does it cost the plant? [\[Nutrition in Plants\]](#)

- (A) it saves water by losing less through the pores, but it slows photosynthesis as less carbon dioxide enters
- (B) it saves carbon dioxide, but loses more water
- (C) it cools the plant by letting water out faster
- (D) it gains nitrogen, but loses oxygen
- (E) it lets out oxygen faster, but takes in no light
- (F) it lets in more water, but loses sunlight
- (G) it speeds photosynthesis, but wastes water

72. A jar has 12 toffees: 4 orange, 5 lemon and 3 mint. One toffee is taken at random. What is the probability that it is NOT lemon? [\[Data Handling\]](#)

- (A) $\frac{5}{12}$
- (B) $\frac{7}{5}$
- (C) $\frac{7}{12}$
- (D) $\frac{1}{2}$
- (E) $\frac{5}{7}$
- (F) $\frac{1}{12}$
- (G) $\frac{2}{3}$

73. A bath needs water at 40 C. You have 30 litres of hot water at 60 C and want to cool it by adding cold water at 20 C. Mixing equal masses, the final temperature is the average of the two. How many litres of cold water give exactly 40 C? [\[Heat\]](#)

- (A) 40 litres
- (B) 15 litres
- (C) 60 litres
- (D) 20 litres
- (E) 30 litres
- (F) 45 litres

74. The sum of two rational numbers is $-\frac{5}{6}$. If one of them is $-\frac{2}{3}$, what is the other? [\[Rational Numbers\]](#)

- (A) $\frac{7}{6}$
- (B) $-\frac{7}{6}$
- (C) $-\frac{1}{3}$
- (D) $\frac{1}{6}$
- (E) $\frac{3}{2}$
- (F) $-\frac{3}{2}$
- (G) $-\frac{1}{6}$

75. The food a plant makes is glucose, but plants store it mainly as a different substance that is not sweet and does not dissolve easily, so it stays put in storage organs. What is this stored form, and which simple test detects it? [\[Nutrition in Plants\]](#)

- (A) sugar, detected by lime water turning milky
- (B) starch, detected by iodine turning blue-black
- (C) chlorophyll, detected by iodine turning blue-black
- (D) starch, detected by alcohol turning green
- (E) starch, detected by lime water turning milky
- (F) protein, detected by iodine turning blue-black
- (G) fat, detected by iodine turning red

76. The mean of five numbers is 14. If each of the five numbers is increased by 3, what is the new mean?

[\[Data Handling\]](#)

- (A) 14
- (B) 14.6
- (C) 17
- (D) 20
- (E) 70
- (F) 15.5
- (G) 85

77. A weather station logs the temperature every 6 hours starting at midnight: 18 C, 24 C, 31 C, 21 C. What is the mean temperature for the day, and what is the range between hottest and coldest? [\[Heat\]](#)

- (A) mean 23.5 C, range 31 C
- (B) mean 23.5 C, range 10 C
- (C) mean 23 C, range 7 C
- (D) mean 22 C, range 13 C
- (E) mean 23.5 C, range 13 C
- (F) mean 94 C, range 13 C

78. By what number should $-\frac{3}{7}$ be divided to give the result -1 ? [\[Rational Numbers\]](#)

- (A) $-\frac{3}{7}$
- (B) -1
- (C) $\frac{3}{7}$
- (D) $\frac{3}{49}$
- (E) $\frac{7}{3}$
- (F) $-\frac{7}{3}$

79. A plant in a sealed glass box is left in complete darkness for two days, then its leaf is tested for starch. What result is expected, and why is this 'de-starching' done before many photosynthesis experiments?

[\[Nutrition in Plants\]](#)

- (A) no starch is found; the leaf has no chlorophyll left
- (B) the leaf turns blue-black at once from the dark
- (C) lots of starch is found; darkness makes extra starch
- (D) plenty of starch is found; this proves light is not needed
- (E) no starch is found; darkness stops photosynthesis and uses up stored starch, giving a clean starting point
- (F) starch doubles in the dark, so the test fails

80. A dartboard's chance of hitting the bullseye is $\frac{1}{8}$ on each throw. Out of 240 throws, about how many bullseyes would you expect? [\[Data Handling\]](#)

- (A) 30
- (B) 240
- (C) 32
- (D) 8
- (E) 15
- (F) 120
- (G) 60

81. Three identical cups of tea are left to cool: cup A is open, cup B has a lid, cup C has a lid and is wrapped in wool. After 10 minutes which is hottest, and which way of losing heat does each extra cover mainly block? [\[Heat\]](#)

- (A) cup C is hottest; the lid makes heat and the wool reflects it
- (B) cup A is hottest; covers make tea cool faster
- (C) cup A is hottest; wool is a good conductor
- (D) cup C is hottest; both covers block only radiation
- (E) cup B is hottest; the lid adds heat
- (F) cup C is hottest; the lid blocks convection from the surface and the wool slows conduction
- (G) all three are equally hot; covers do nothing

82. Compute $(-\frac{3}{5}) + (-\frac{3}{5}) + (-\frac{3}{5}) + (-\frac{3}{5})$. Express the answer in lowest terms. [\[Rational Numbers\]](#)

- (A) $-\frac{12}{20}$
- (B) $\frac{12}{5}$
- (C) $-\frac{9}{5}$
- (D) $-\frac{12}{5}$
- (E) $-\frac{3}{20}$
- (F) $-\frac{2}{5}$
- (G) $-\frac{3}{5}$

83. A pond has 600 algae cells at the start. With plenty of sunlight their number triples every day. How many algae cells are there after 2 full days? [\[Nutrition in Plants\]](#)

- (A) 1200 cells
- (B) 16200 cells
- (C) 5400 cells
- (D) 3000 cells
- (E) 1800 cells
- (F) 3600 cells
- (G) 2400 cells

84. Six readings have a mean of 50. Five of them are 48, 52, 47, 55 and 50. Find the sixth reading, and state whether it is above or below the mean. [\[Data Handling\]](#)

- (A) 48, which is above the mean
- (B) 300, which is above the mean
- (C) 2, which is below the mean
- (D) 48, which is below the mean
- (E) 50, which equals the mean
- (F) 52, which is below the mean
- (G) 52, which is above the mean

85. A metal scale is exactly 100 cm long at 20 C. When it is heated it expands by 0.2 mm for every 10 C rise. How long is the scale at 70 C? [\[Heat\]](#)

- (A) 100.01 cm
- (B) 102 cm
- (C) 110 cm
- (D) 100.5 cm
- (E) 100.1 cm
- (F) 100.2 cm

86. A number n satisfies $n - (-3/4) = 1/8$. What is n ? [\[Rational Numbers\]](#)

- (A) $5/8$
- (B) $7/8$
- (C) $-1/8$
- (D) $-7/8$
- (E) $-11/8$
- (F) $11/8$
- (G) $-5/8$

87. Insectivorous plants, parasites like *Cuscuta*, and saprotrophic fungi all get food in special ways. Which one of these still carries out photosynthesis to make its own glucose, and only uses its special trick for extra nutrients? [\[Nutrition in Plants\]](#)

- (A) the insectivorous plant - it has green leaves and only traps insects for nitrogen
- (B) *Cuscuta* - it makes glucose and traps insects too
- (C) *Cuscuta* - it is green and photosynthesises fully
- (D) all three photosynthesise normally
- (E) none of them can photosynthesise
- (F) the insectivorous plant - it cannot photosynthesise at all
- (G) the saprotrophic fungus - it photosynthesises on dead matter

88. In a survey, 7 people chose tea, 5 chose coffee, and the rest of the 20 people chose milk. If one person is picked at random, what is the probability they chose milk? [\[Data Handling\]](#)

- (A) $\frac{2}{3}$
- (B) $\frac{7}{20}$
- (C) $\frac{2}{5}$
- (D) $\frac{3}{5}$
- (E) $\frac{1}{2}$
- (F) $\frac{8}{20}$ unsimplified only
- (G) $\frac{1}{4}$

89. Boiling water is at 100 C and ice is melting at 0 C. A scientist marks the midpoint of this range as a reference. If a sample sits one quarter of the way up from the ice point, what is its temperature? [\[Heat\]](#)

- (A) 75 C
- (B) 50 C
- (C) 25 C
- (D) 20 C
- (E) 40 C
- (F) 30 C
- (G) 12.5 C

90. Which rational number is its own reciprocal as well as being negative? [\[Rational Numbers\]](#)

- (A) 2
- (B) there is none
- (C) 0
- (D) 1
- (E) -2
- (F) -1
- (G) $-\frac{1}{2}$

91. A leaf takes in carbon dioxide through its stomata. If the air around the leaf has more carbon dioxide than usual (and light and water are plenty), what happens to the rate of photosynthesis, up to a point?

[\[Nutrition in Plants\]](#)

- (A) it slows down, because carbon dioxide is harmful
- (B) it speeds up, because more raw material is available to build glucose
- (C) it stops at once, because too much gas blocks the leaf
- (D) it doubles instantly no matter how much light there is
- (E) it stays exactly the same, because CO₂ has no effect
- (F) it slows down, because CO₂ closes the stomata
- (G) it speeds up, because CO₂ supplies the energy

92. A student's marks in five subjects are 72, 64, 88, 76 and x. The mean must be at least 75 to win a prize. What is the smallest whole-number value of x that wins the prize? [\[Data Handling\]](#)

- (A) 70
- (B) 72
- (C) 74
- (D) 375
- (E) 73
- (F) 76
- (G) 75

93. A patient's temperature is taken every hour for four hours: 38.5 C, 39.0 C, 38.0 C, 37.5 C. By how much did the fever fall from its highest reading to the last reading, and was the trend overall up or down?

[Heat]

- (A) fell 2.0 C; the trend was downward
- (B) rose 1.5 C; the trend was upward
- (C) fell 1.5 C; the trend was upward
- (D) fell 1.5 C; there was no trend
- (E) fell 1.0 C; the trend was downward
- (F) fell 1.5 C; the trend was downward
- (G) fell 0.5 C; the trend was downward

94. Three friends share a debt. The debt is written as the rational number $-27/4$ rupees-thousand and split equally. What signed amount does each friend carry? [Rational Numbers]

- (A) $27/12$
- (B) $9/4$
- (C) $-81/4$
- (D) $-9/4$
- (E) $-9/12$
- (F) $-3/4$

95. A balanced way to write photosynthesis is: carbon dioxide + water, using sunlight trapped by chlorophyll, gives glucose + oxygen. From this, where does the oxygen we breathe out by plants by day come from, and where does the glucose go? [Nutrition in Plants]

- (A) the oxygen is absorbed by roots; glucose leaves through stomata
- (B) the oxygen is released into the air; the glucose is used for energy or stored as starch
- (C) the glucose becomes sunlight again; oxygen makes water
- (D) both oxygen and glucose stay locked inside the chlorophyll
- (E) the oxygen is used to make chlorophyll; glucose is wasted
- (F) the oxygen and glucose are both breathed out as gas
- (G) the oxygen is stored as starch; the glucose escapes into the air

96. Two dice are rolled and their numbers added. Out of the 36 equally likely outcomes, how many give a sum of 7, and what is the probability of a sum of 7? [Data Handling]

- (A) 6 outcomes, probability $6/6$
- (B) 11 outcomes, probability $11/36$
- (C) 5 outcomes, probability $5/36$
- (D) 1 outcome, probability $1/36$
- (E) 6 outcomes, probability $1/6$
- (F) 6 outcomes, probability $1/12$
- (G) 7 outcomes, probability $7/36$

97. A solar water heater raises the temperature of 40 litres of water from 22 C to 58 C on a sunny day. Another day it manages only a 24 C rise on the same 40 litres. By what fraction of the first day's rise did the heating fall on the second day? [\[Heat\]](#)

- (A) by $\frac{1}{3}$ of the first day's rise
- (B) by $\frac{1}{6}$ of the first day's rise
- (C) by $\frac{12}{36}$ is not it
- (D) by $\frac{2}{3}$ of the first day's rise
- (E) by $\frac{1}{2}$ of the first day's rise
- (F) by $\frac{24}{58}$ of the first day's rise

98. Evaluate $(\frac{2}{3} - \frac{5}{6}) \times (-\frac{9}{2})$. Give the answer in lowest terms. [\[Rational Numbers\]](#)

- (A) $-\frac{3}{4}$
- (B) $-\frac{1}{6}$
- (C) $\frac{3}{4}$
- (D) $-\frac{9}{4}$
- (E) $\frac{1}{6}$
- (F) $\frac{3}{2}$

99. A gardener wants the fastest plant growth. She can raise the light, the carbon dioxide, or the temperature, but only one factor is below what the plant needs while the others are already plenty. What is this slowest factor called, and what happens if she raises the others instead? [\[Nutrition in Plants\]](#)

- (A) there is no such factor; all three matter equally always
- (B) it is the limiting factor; raising the others has little effect until the slowest one is raised
- (C) it is the waste factor; raising the others stops growth
- (D) it is the limiting factor; raising the others harms the plant
- (E) it is the storage factor; raising the others doubles growth
- (F) it is the resting factor; the plant ignores all changes

100. The mean weight of 8 parcels is 5 kg. Two more parcels, each weighing 9.5 kg, are added. What is the new mean weight of all ten parcels? [\[Data Handling\]](#)

- (A) 5.5 kg
- (B) 5.45 kg
- (C) 6 kg
- (D) 59 kg
- (E) 5.9 kg
- (F) 6.5 kg
- (G) 7.25 kg